

ARAVINDHAN AK

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github.com/ARAVIND56722 | Portfolio | LeetCode: 140+ Problems Solved

Objective

Final-year AI & Data Science undergraduate with hands-on experience across full-stack development, blockchain, and applied AI. Skilled at translating problem statements into working, well-structured software through strong fundamentals in data structures, algorithms, and object-oriented design. Eager to contribute to real-world engineering teams while continuing to grow as a versatile, detail-oriented developer.

Technical Skills

Core CS Fundamentals	Data Structures & Algorithms
Programming Languages	Java, Python
Databases	MongoDB
AI / NLP	Natural Language Processing, Deep Learning, Machine Learning, RAG, LLMs
Data Analytics	Excel, Power BI, Tableau, Data Visualization
Blockchain	Solidity, Ethereum, Web3.py, Smart Contracts, MetaMask, Ganache, Hard-Hat
Tools & Platforms	Remix IDE, Sepolia, Etherscan, Git, GitHub, Vercel, Render

Projects

URL Shortener Pro – Smart Link Management Platform	Jun 2026
<i>MERN Stack (React, Node.js, Express, MongoDB)</i>	<i>Live Demo</i>

- Built a full-stack SaaS link-management platform with JWT authentication, bcrypt password hashing, and protected routes.
- Implemented custom aliases, QR code generation, bulk URL shortening via CSV upload, and link expiry.
- Developed a real-time analytics dashboard tracking clicks, browser/device/OS data, and daily traffic trends using Recharts.
- Secured the API with rate limiting, Helmet, CORS, and input validation.

NGO Management System	Dec 2025
<i>Full-Stack Web Development, CRUD, Database Design</i>	<i>Live Demo</i>

- Built a web-based system to manage donors, volunteers, and events for NGOs.
- Implemented complete CRUD operations for efficient data handling and record tracking.
- Improved operational transparency and record management through structured data workflows.

Citizen Issue Reporting & Tracking System	Sep 2025
<i>Full-Stack Web Development, Role-Based Access Control</i>	<i>Live Demo</i>

- Developed a web platform for citizens to report and track public issues in real time.
- Implemented role-based access control separating citizen and authority workflows.
- Improved issue monitoring, response tracking, and civic service transparency.

ETK Energy Sharing Dashboard	Aug 2025
<i>Web3.py, ERC-20, Streamlit, Solidity</i>	<i>GitHub Repository</i>

- Developed a blockchain-enabled peer-to-peer energy trading dashboard.
- Integrated an ERC-20 smart contract with a Streamlit front end using Web3.py on the Sepolia testnet.
- Implemented ETK token transfers, wallet balance tracking, and live transaction monitoring.

Autonomous AI Research Agent	Mar 2026
<i>React (Vite), FastAPI, Python, RAG, LLM Reasoning</i>	<i>GitHub Repository</i>

- Built a full-stack GenAI application to search, read, and reason over real research papers via the arXiv

API.

- Applied Retrieval-Augmented Generation (RAG) and agentic multi-step workflows to generate topic insights.
- Implemented a context-based Q&A feature allowing users to ask questions about a selected paper.
- Designed a clean, user-friendly UI as an alternative to typical dark AI interfaces.

Education

B.Tech, Artificial Intelligence & Data Science

2023 – 2027

Dr. N.G.P. Institute of Technology

CGPA: 8/10 (up to Sem 6)

Higher Secondary Education

2022 – 2023

G.R.G. Matriculation Higher Secondary School

510/600 (Cutoff: 160.5/200)

Internship / Professional Experience

Data Analyst Intern – I-BACUS-TECH

15 Days, 2025

- Developed practical skills in data visualization and reporting using Power BI.
- Strengthened data analytics fundamentals under a short, time-bound project cycle.

Hackathons

- **Winner** – International Hackathon 360° for "Transparent Blockchain Ledger with Microgrid Energy Sharing" – KPR Institute of Engineering and Technology (2026)
- **Finalist** – National Level Hackathon CRYPTERA 2026, "NGO Management System" – Coimbatore Institute of Technology (2025)
- **Participant** – 24-Hour Hackathon, "Citizen Issue Reporting & Tracking System" – KPR Institute of Engineering and Technology (2025)
- **Participant** – Hackathon, "Blockchain-Based Supply Chain Management" – SNS Institute of Technology (2026)

Certifications

- Privacy and Security in Online Social Media – NPTEL (2025)
- Data Analytics – NoviTech R&D Private Limited (2025)
- Human Computer Interaction – NPTEL (2026)
- Generative AI with AWS – Analytics Vidhya (2026)

Co-Curricular Activities

- Honors in Blockchain Technology – Dr. N.G.P. Institute of Technology (2026)
- AI/4.0 Solution Expo – Dr. N.G.P. Institute of Technology, Mar 6, 2026
- Project Expo 2026, "Food Rescue Logistic Platform" – Dr. N.G.P. Institute of Technology, Feb 10–11, 2026
- Presented paper at ICRIT 2026 – Grace College of Engineering, Thoothukudi, Apr 20–21, 2026
- Presented paper at NCATSM 2026 – Karpagam Academy of Higher Education, Coimbatore, Mar 13, 2026

Languages Known

English | Tamil